WebTech

WEEK 1

992401040105 Ananya Soni

Q1

```html
<!DOCTYPE html>

<html>

  <head>

    <title> Profile</title>

  </head>

  <body>

    <h1 align="center"> Ananya Soni </h1>

    <h2 align="center"> B.TECH - Information Technology</h2>

    <p align="center"><font size="5px">

      I am Ananya Soni, a 2nd year student at <a
href="https://www.jiit.ac.in/"><b>Jaypee Institute Of Information Technology</a>,
Noida 201301 </b><br><hr>

    </font>

    </p>

    <p align="right">I have interests in coding, game development, cybersecurity and
many other creative fields <br>

      I have created few projects such as Web Scraper which <br>

      Other than techincal I created a 2D story driven game with my friends using
GODOT.

    </p>

  </body>

</html>
```

Ananya Soni

B.TECH - Information Technology

I am Ananya Soni, a 2nd year student at [Jaypee Institute Of Information Technology](), **Noida 201301**

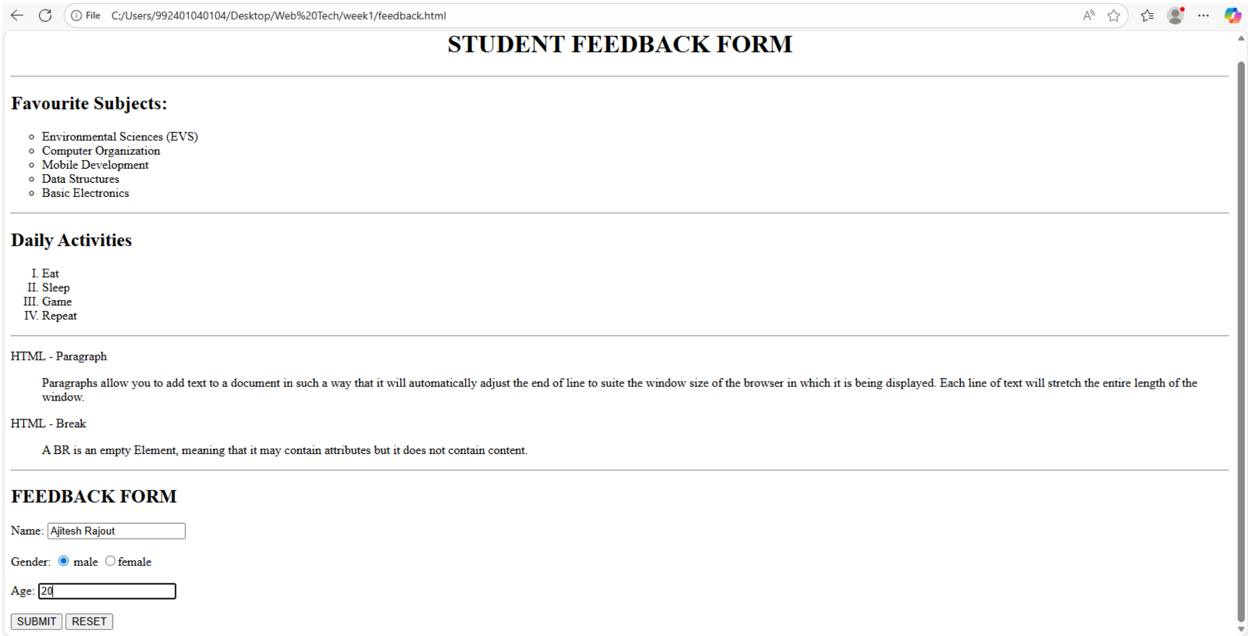
I have interests in coding, game development, cybersecurity and many other creative fields
I have created few projects such as Web Scraper which
Other than techincal I created a 2D story driven game with my friends using GODOT.

Q2

```
<!DOCTYPE html>
<html>
  <head>
    <title> Student Feedback Page </title>
  </head>
  <body>
    <h1 align="center"> STUDENT FEEDBACK FORM</h1>
    <hr>
    <h2> Favourite Subjects:</h2>
    <ul type="circle">
      <li > Environmental Sciences (EVS)</li>
      <li> Computer Organization </li>
      <li> Mobile Development</li>
      <li> Data Structures</li>
      <li> Basic Electronics</li>
    </ul>
    <hr>
    <h2> Daily Activities </h2>
    <ol type="I">
      <li> Eat</li>
```

```
    <li> Sleep</li>
    <li> Game</li>
    <li> Repeat</li>
  </ol>
  <hr>
  <dl>
    <dt>HTML - Paragraph</dt>
    <dd>
      <p>Paragraphs allow you to add text to a document in such a way that it will automatically adjust the end of line to suite the window size of the browser in which it is being displayed. Each line of text will stretch the entire length of the window.</p>
    </dd>
    <dt>HTML - Break</dt>
    <dd>
      <p>A BR is an empty Element, meaning that it may contain attributes but it does not contain content.</p>
    </dd>
  </dl>
  <hr>

  <H2>FEEDBACK FORM</H2>
  <form method="post" action="">
    Name: <input type="text" size="20">
    <br>
    <br>
    Gender: <input type="radio" name="gender" value="m"> male
    <input type="radio" name="gender" value="f">female
    <br>
    <br>
```

```
            Age: <input type="text" name="age" size="20">
            <br>
            <br>
            <input type="submit" value="SUBMIT">
            <input type="reset" value="RESET">
        </form>
    </body>
</html>
```



WebTech

WEEK 2

992401040105 Ananya Soni

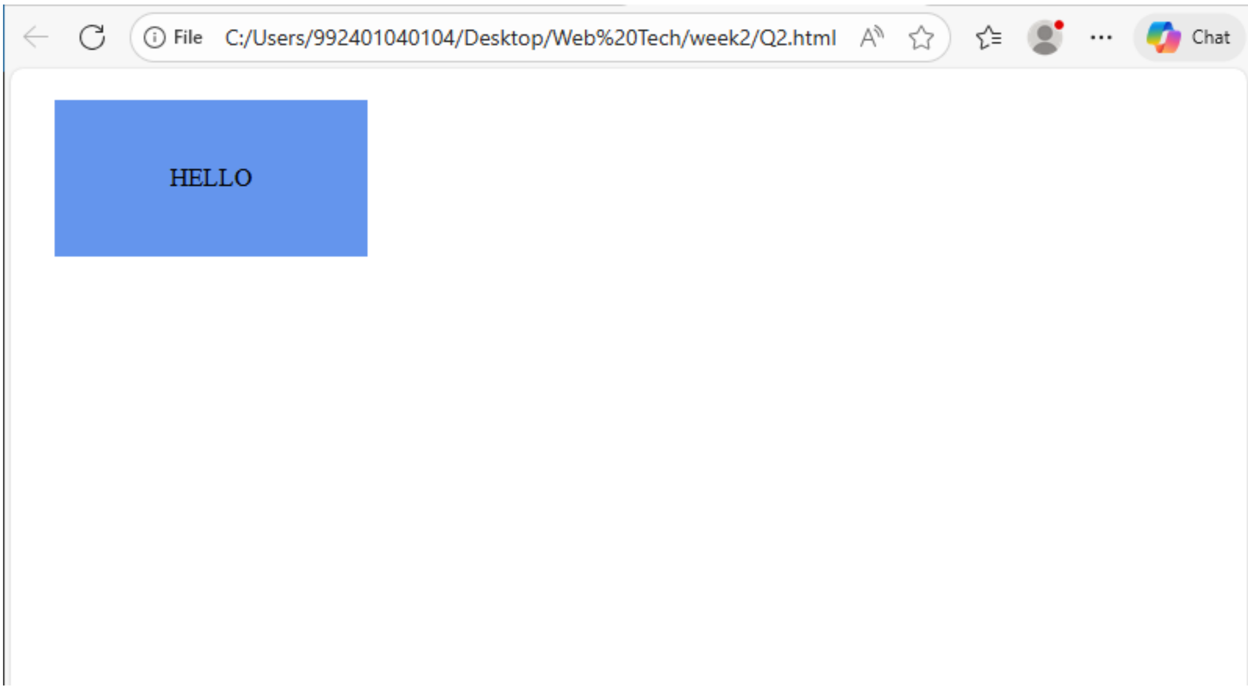
Q1

```html
<!DOCTYPE html>
<html>
  <head>
    <title>
      W2Q1
    </title>
    <style>
      h1{
        color: purple;
        font-size: 50px ;
      }
      p{
        color: olive;
        background-color: lightgreen;
        padding: 50px;
      }
    </style>
  </head>
  <body>
    <h1>This Is A Heading</h1>
    <p>This is a PARAGRAPH</p>
  </body>
</html>
```

Q2

```
<!DOCTYPE html>
<html>
  <head>
    <title>
      W2Q2
    </title>
    <style>
      #d{
        width: 200px;
        height: 100px;
        background-color: cornflowerblue;
        border: 2px;
        margin: 20px;
        text-align: center;
        align-content: center;
      }
    </style>
```

```
  </head>
  <body>
    <div id="d">
      <p>HELLO</p>
    </div>
  </body>
</html>
```



Q3

```
<!DOCTYPE html>
<html>
  <head>
    <title>
      W2Q3
```

```
    </title>
    <style>
      button{
        background-color: khaki;
        color: darkkhaki;
        border: none;
        padding: 50px;


      }
      button:hover{
        background-color: blue;
      }
    </style>
  </head>
  <body>
    <button>
      hello
    </button>
  </body>
</html>
```

Before Hover:


File
C:/Users/992401040104/Desktop/Web%20Tech/week2/Q3.html
Chat
hello

After Hover:


File
C:/Users/992401040104/Desktop/Web%20Tech/week2/Q3.html
Chat
hello

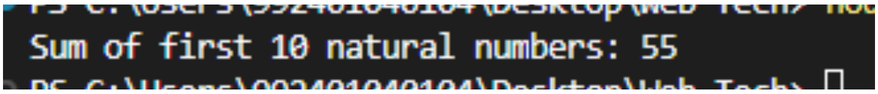
WebTech

WEEK 3

992401040105 Ananya Soni

Q1

```
var sum=0;
for(var i=1;i<11;i++){
  sum+=i;
}
console.log("Sum of first 10 natural numbers: "+sum);
```



Q2

```
function max(a,b){
  if(a>b){
    return a;
  }
  else if(b>a){
    return b;
  }
  else{
    return "Both are same";
  }
}

var largest=max(15,10);
console.log("Largest number: " + largest);
```

Q3

```
const vowel=['a','e','i','o','u','A','E','I','O','U',' '];
```

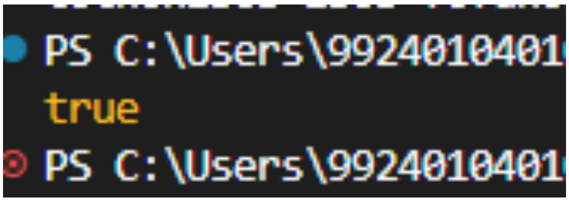
```
function translate(s){
  var ans='';
  for(var i=0;i<s.length;i++){
    if(!vowel.includes(s[i])){
      ans=ans+s[i]+'o'+s[i];
    }
    else{
      ans+=s[i];
    }
  }
  return ans;
}
var s="this is fun";
var f=translate(s);
console.log("Translation of '"+s+"' is:\n"+f);
```

```
Translation of 'this is fun' is:
tothohisos isos fofunon
```

Q4

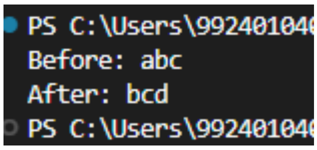
```
var a=100;
var b=200;

function check(a,b){
  if(a==100 || b==100 || a+b==100){
    return true;
  }
  else{
    return false;
  }
```

```
}
console.log(check(a,b));
```



Q5

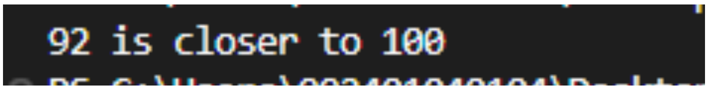
```
function replace(s){
  var result=''
  for(var i=0;i<s.length;i++){
    let newChar = String.fromCharCode(s.charCodeAt(i) + 1);
    result += newChar;
  }
  return result;
}

var s="abc";
console.log("Before: "+s);
s=replace(s);
console.log("After: "+s);
```



Q6

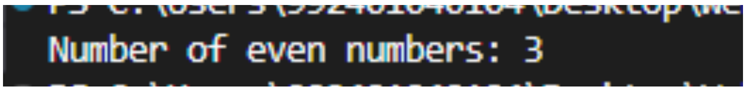
```
var a=86;
var b=92;

function close(a,b){
  if(Math.abs(100-a)>Math.abs(100-b)){
    return b;
  }
  else if(Math.abs(100-a)<Math.abs(100-b)){
```

```
    return a;
  }
  else{
    return "Both are same";
  }
}
console.log(close(a,b) +" is closer to 100");
```



Q7

```
var l=[1,2,3,4,5,6];

function even(l){
  var c=0;
  for(var i=0;i<l.length;i++){
    if (l[i]%2==0){
      c++;
    }
  }
  return c;
}

console.log("Number of even numbers: "+even(l));
```



Q8

```
function coins(a){
  let ans=[];
```

```
  while(a!=0){
    if(a>=25){
      ans.push(25);
      a-=25;
      //console.log(25);
    }
    else if(a>=10){
      ans.push(10);
      a-=10;
    }
    else if(a>=5){
      ans.push(5);
      a-=5;
    }
    else if(a>=2){
      ans.push(2);
      a-=2;
    }
    else if(a>=1){
      ans.push(1);
      a-=1;
    }
  }
  return ans;
}

var a=46;
console.log(coins(a));
```

```
PS C:\Users\99240104016
[ 25, 10, 10, 1 ]
PS C:\Users\99240104016
```

Q9

```
function stutter(a){
  var ans=[];
  for(var i=0;i<a.length;i++){
    ans.push(a[i]);
    ans.push(a[i]);
  }
  return ans;
}

var a=[1,2,3];
console.log("Before stutter:" +a);
console.log("After stutter:" +stutter(a,4));
```

```
1.js"
Before stutter:1,2,3
After stutter:1,1,2,2,3,3
PS C:\Users\992401040104\Desktop\W
```

Q10

```
function cycle(a,n){
  if(n<=0){
    return a;
  }
  var ans=[];
  for(var i=n;i<a.length;i++){
    //console.log(a[i]);
```

```
        ans.push(a[i]);
    }
    for(var i=0;i<n;i++){
        ans.push(a[i]);
    }
    return ans;

}

var a=[1,2,3,4,5,6];
console.log("Before rearranging:" +a);
console.log("After rearranging:" +cycle(a,4));
```

```
1.js"
Before rearranging:1,2,3,4,5,6
After rearranging:5,6,1,2,3,4
```

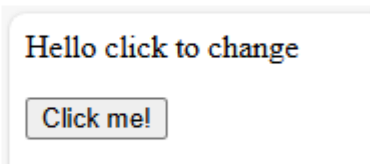
WEEK 4

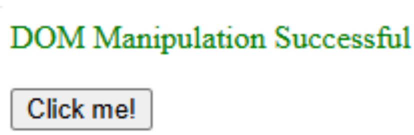
ANANYA SONI 992401040105

1)

```
<html>
  <head>
    <title>Q1</title>
  </head>
  <body>
    <p id="change">Hello click to change</p>
    <button onclick="func()">Click me!</button>
    <script>
      function func(){
        document.getElementById("change").innerHTML="DOM Manipulation Successful";
        document.getElementById("change").style.color="green";
      }
    </script>
  </body>
</html>
```

Output:



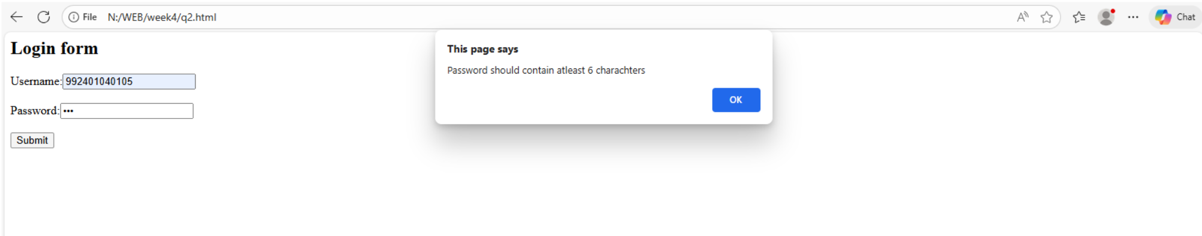


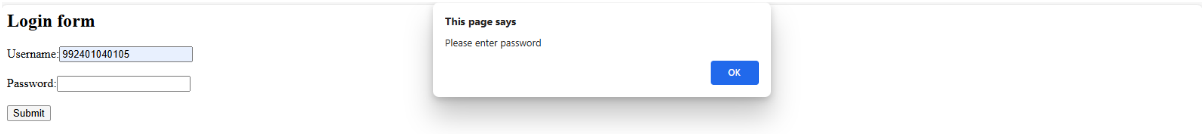
2)

```
<html>
    <head><title>Q2</title></head>
    <body>
        <h2>Login form</h2>
```

```
    <form method="post" action="" onsubmit="return validateform()">
            Username:<input type="Text" id="username"><br><br>
            Password:<input type="password" id="password"><br><br>
            <button type="submit">Submit</button>
        </form>
        <script>
            function validateform(){
                var u=document.getElementById('username').value;
                var p=document.getElementById('password').value;
                if(u==""){
                    alert("Please enter username");
                    return false;
                }
                if(p==""){
                    alert("Please enter password");
                    return false;
                }
                if(p.length<6){
                    alert("Password should contain atleast 6 charachters");
                    return false;
                }
                alert("All data fields are valid");
                return true;
            }
        </script>
    </body>
</html>
```

Output:





3)

```
<html>
    <head><title>Q3</title></head>
    <body>
        <img src="winter.jpg" id="ig">
        <button id="change" onclick="picture()">Change season</button>
        <button id="res" onclick="resize()">Resize</button>
        <script>
            function picture(){
                var img=document.getElementById("ig");
                img.src="summer.jpg";
            }
            function resize(){
                var img=document.getElementById("ig");
                img.style.width=1000+"px";
                img.style.width=1000+"px";
            }
        </script>
    </body>
</html>
```

Output:

Change season Resize

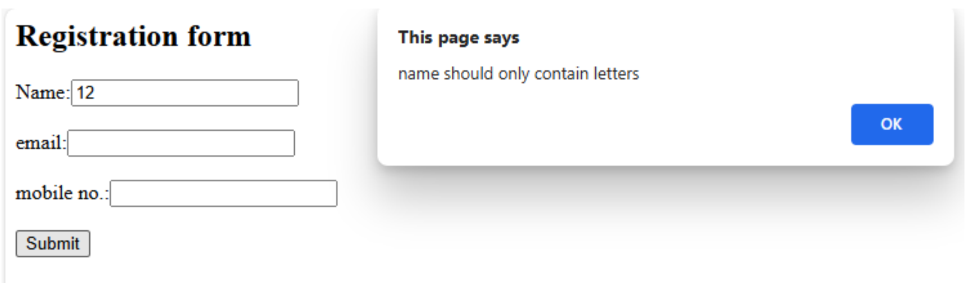
Change season Resize

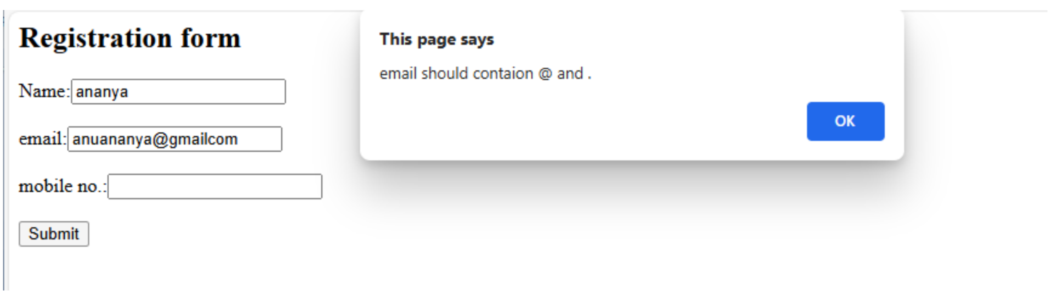
4)

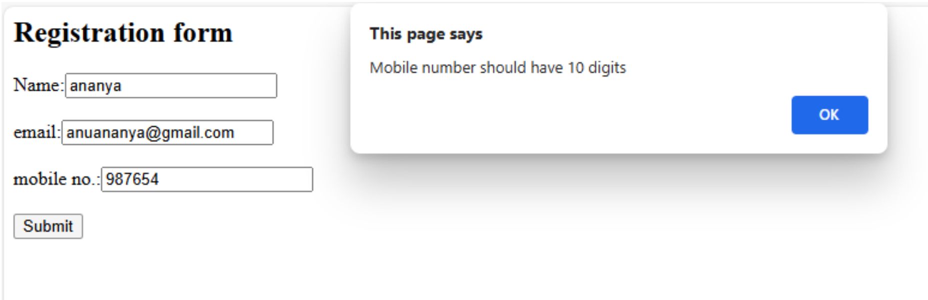
```
<html>
    <head><title>Q4</title></head>
    <body>
        <h2>Registration form</h2>
        <form method="post" action="" onsubmit="return validateform()">
            Name:<input type="Text" id="name"><br><br>
            email:<input type="email" id="email"><br><br>
            mobile no.:<input type="text" id="mobile"><br><br>
            <button type="submit">Submit</button>
        </form>
        <script>
            function validateform(){
                var n=document.getElementById("name").value;
                var e=document.getElementById("email").value;
                var m=document.getElementById("mobile").value;
                for(let i=0;i<n.length;i++){
                    let code=n.charCodeAt(i);
                    if(!(code >= 65 && code <= 90) && !(code >= 97 && code <= 122)){
                        alert("name should only contain letters");
```

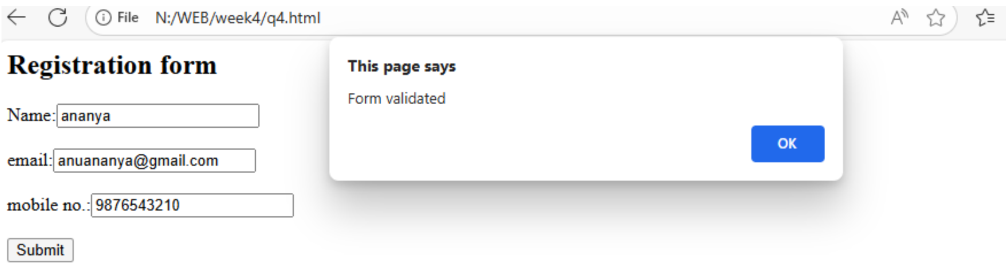
```
                    return false;
                }
            }
            if(!e.includes('@') || !e.includes('.')){
                alert("email should contaion @ and .");
                return false;
            }
            if(m.length!=10){
                alert("Mobile number should have 10 digits");
                return false;
            }
            alert("Form validated");
            return true;
        }
    </script>
  </body>
</html>
```

Output:







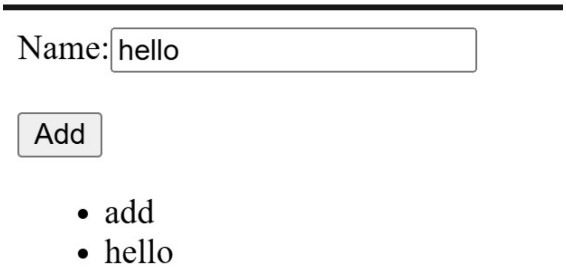


5)

```
<html>
  <head><title>Q5</title></head>
  <body>
    Name:<input type="text" id="ty"><br><br>
    <button id="add" onclick="add()">Add</button>
    <ul id="lit"></ul>
    <script>
      function add(){
        let a= document.getElementById('ty').value;
        if(a===''){
          return;
        }
        let t1=document.getElementById('lit');
        let t2="<li>"+a+"</li>"
        t1.innerHTML+=t2;
      }
```

```
    </script>
  </body>
</html>
```

Output:

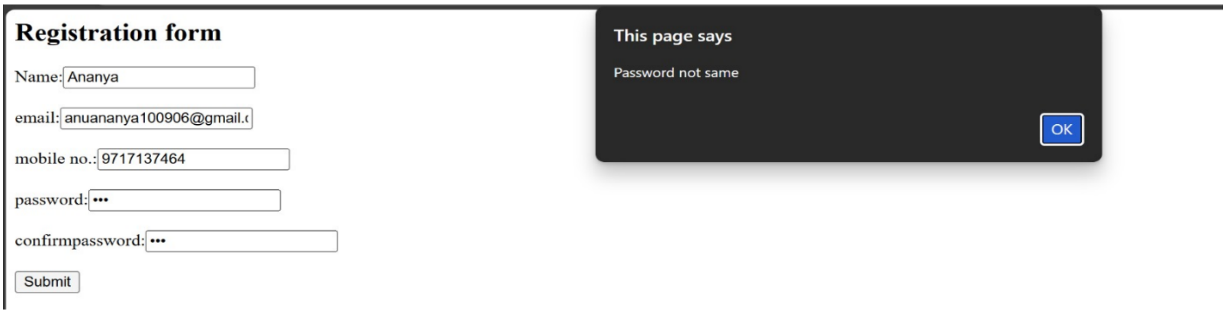


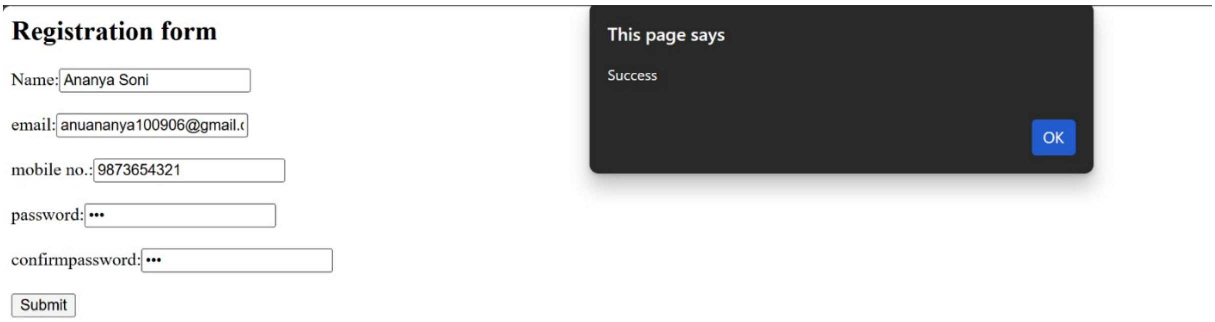
6)

```
<html>
  <head><title>Q4</title></head>
  <body>
    <h2>Registration form</h2>
    <form method="post" action="" onsubmit="return validateform()">
      Name:<input type="Text" id="name"><br><br>
      email:<input type="email" id="email"><br><br>
      mobile no.:<input type="text" id="mobile"><br><br>
      password:<input type="password" id="pass"><br><br>
      confirmpassword:<input type="password" id="confirmpass"><br><br>
      <button type="submit">Submit</button>
    </form>
    <script>
      function validateform(){
        var n=document.getElementById("name").value;
        var e=document.getElementById("email").value;
```

```
            var m=document.getElementById("mobile").value;
            var p=document.getElementById("pass").value;
            var cp=document.getElementById("confirmpass").value;
            if(p===cp){
              alert("Success");
            }
            else{
              alert("Password not same");
            }
            for(let i=0;i<n.length;i++){
              let code=n.charCodeAt(i);
              if(!(code >= 65 && code <= 90) && !(code >= 97 && code <= 122)){
                alert("name should only contain letters");
                return false;
              }
            }
            if(!e.includes('@') || !e.includes('.')){
              alert("email should contaion @ and .");
              return false;
            }
            if(m.length!=10){
              alert("Mobile number should have 10 digits");
              return false;
            }
            alert("Form validated");
            return true;
          }
        </script>
    </body>
</html>
```

Output:



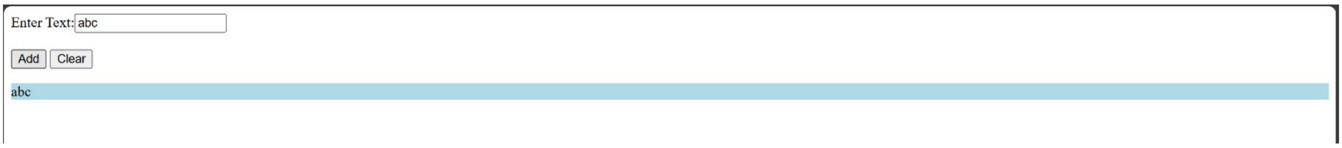


7)

```
<html>
<head>
  <title>Text Add & Clear</title>
</head>
<body>
  Enter Text:<input type="text" id="txt"><br><br>
  <button onclick="addText()">Add</button>
  <button onclick="clearText()">Clear</button>
  <p id="para"></p>
  <script>
    function addText(){
      let t = document.getElementById("txt").value;
      let p = document.getElementById("para");

      p.innerHTML = t;
      p.style.backgroundColor = "lightblue";
```

```
        }
        function clearText(){
            document.getElementById("para").innerHTML = "";
            document.getElementById("para").style.backgroundColor = "";
            document.getElementById("txt").value = "";
        }
    </script>
</body>
</html>
```

Output:



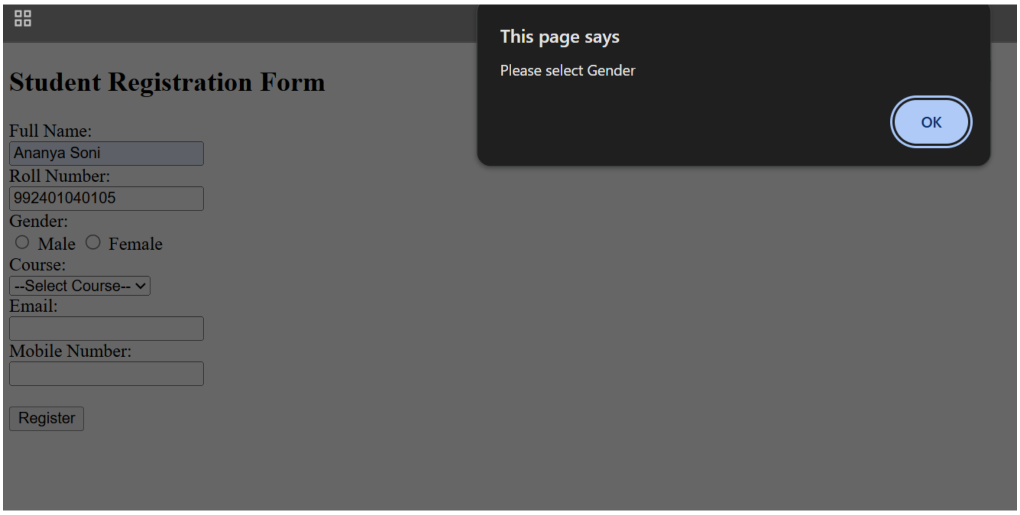
8)

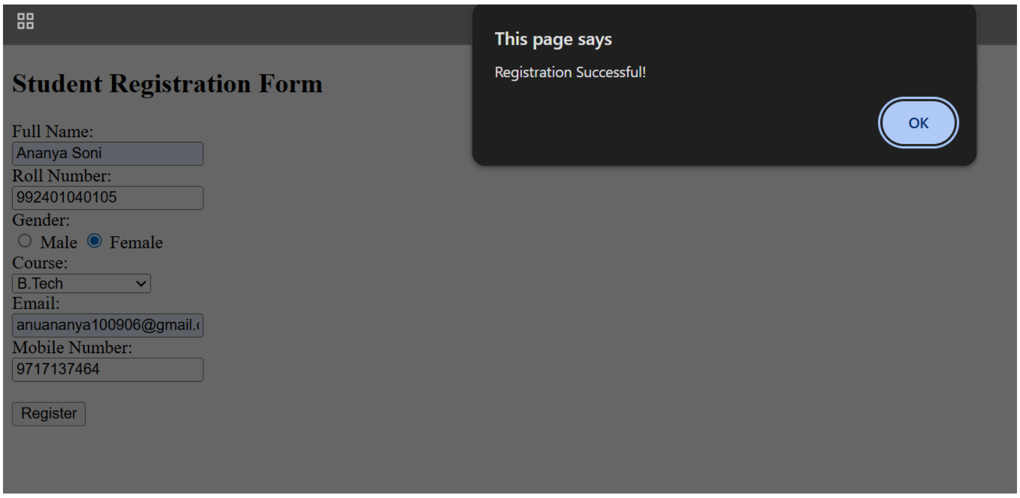
```
<!DOCTYPE html>
<html>
<head>
    <title>Student Registration</title>
</head>
<body>

<h2>Student Registration Form</h2>

<form onsubmit="return validate()">

    Full Name:<br>
    <input type="text" id="name"><br>
```

```
    Roll Number:<br>
    <input type="text" id="roll"><br>

    Gender:<br>
    <input type="radio" name="gender"> Male
    <input type="radio" name="gender"> Female<br>

    Course:<br>
    <select id="course">
        <option value="">--Select Course--</option>
        <option value="BTech">B.Tech</option>
        <option value="BCA">BCA</option>
        <option value="MCA">MCA</option>
    </select><br>

    Email:<br>
    <input type="text" id="email"><br>

    Mobile Number:<br>
    <input type="text" id="mobile"><br><br>

    <input type="submit" value="Register">

</form>

<script>
function validate() {

    var a = document.getElementById("name").value;
    var b = document.getElementById("roll").value;
```

```
var c = document.getElementById("email").value;
var d = document.getElementById("mobile").value;
var e = document.getElementById("course").value;

var g = document.querySelector('input[name="gender"]:checked');

if (a === "") {
   alert("Full Name is required");
   return false;
}

if (b === "") {
   alert("Roll Number is required");
   return false;
}

if (g === null) {
   alert("Please select Gender");
   return false;
}

if (e === "") {
   alert("Please select Course");
   return false;
}

if (c === "") {
   alert("Email is required");
   return false;
}
```

```
    if (d === "") {
        alert("Mobile Number is required");
        return false;
    }

    if (isNaN(d)) {
        alert("Mobile Number must be numeric");
        return false;
    }

    if (d.length != 10) {
        alert("Mobile Number must be 10 digits");
        return false;
    }

    alert("Registration Successful!");
    return true;
}
</script>

</body>
</html>
```



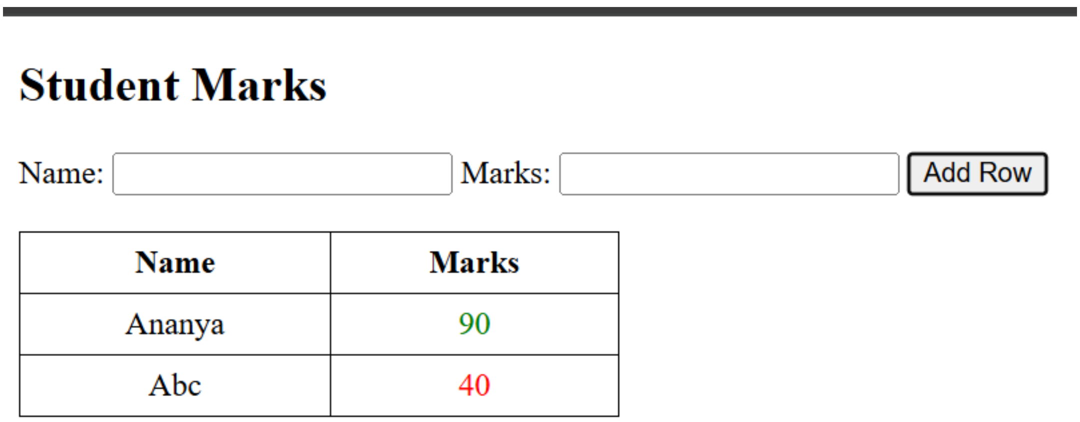


9)

```
<!DOCTYPE html>

<html>

<head>

  <title>Student Marks Table</title>

  <style>

    table {

      border-collapse: collapse;

      width: 300px;

    }

    th, td {

      border: 1px solid black;

      padding: 6px;

      text-align: center;

    }

  </style>

</head>

<body>


<h2>Student Marks</h2>


Name:
```

```
<input type="text" id="name">

Marks:
<input type="text" id="marks">

<button onclick="addRow()">Add Row</button>

<br><br>

<table id="myTable">
  <tr>
    <th>Name</th>
    <th>Marks</th>
  </tr>
</table>

<script>
function addRow() {

  var a = document.getElementById("name").value;
  var b = document.getElementById("marks").value;

  if (a === "" || b === "") {
    alert("Please enter Name and Marks");
    return;
  }

  var c = document.getElementById("myTable");

  var row = c.insertRow();
```

```
    var cell1 = row.insertCell(0);
    var cell2 = row.insertCell(1);

    cell1.innerHTML = a;
    cell2.innerHTML = b;

    if (b > 40) {
      cell2.style.color = "green";
    } else {
      cell2.style.color = "red";
    }

    document.getElementById("name").value = "";
    document.getElementById("marks").value = "";
}
</script>

</body>
</html>
```



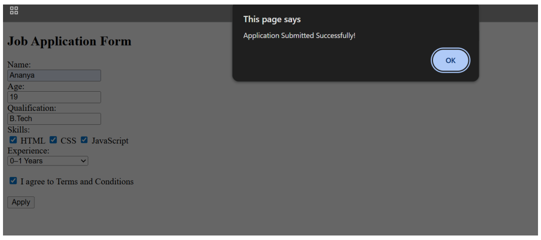
10)

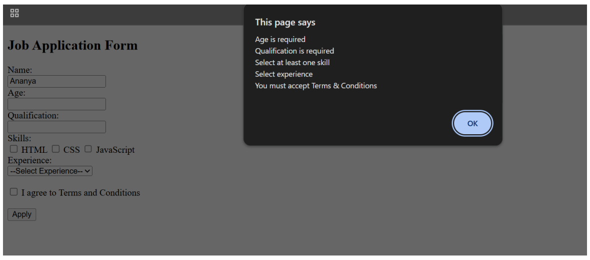
```
<!DOCTYPE html>
```

```html
<html>
<head>
  <title>Job Application Form</title>
</head>
<body>

<h2>Job Application Form</h2>

<form onsubmit="return validateForm()">

  Name:<br>
  <input type="text" id="name"><br>

  Age:<br>
  <input type="text" id="age"><br>

  Qualification:<br>
  <input type="text" id="qualification"><br>

  Skills:<br>
  <input type="checkbox" name="skills" value="HTML"> HTML
  <input type="checkbox" name="skills" value="CSS"> CSS
  <input type="checkbox" name="skills" value="JavaScript"> JavaScript<br>

  Experience:<br>
  <select id="experience">
    <option value="">--Select Experience--</option>
    <option value="0-1">0–1 Years</option>
    <option value="1-3">1–3 Years</option>
    <option value="3+">3+ Years</option>
  </select><br><br>
```

```
    <input type="checkbox" id="terms"> I agree to Terms and Conditions<br><br>

    <input type="submit" value="Apply">

</form>

<script>
function validateForm() {

    let msg = "";

    let name = document.getElementById("name").value.trim();
    let age = document.getElementById("age").value.trim();
    let qualification = document.getElementById("qualification").value.trim();
    let experience = document.getElementById("experience").value;
    let terms = document.getElementById("terms").checked;

    let skills = document.querySelectorAll('input[name="skills"]:checked');

    if (name === "")
        msg += "Name is required\n";

    if (age === "")
        msg += "Age is required\n";
    else if (isNaN(age) || age < 18)
        msg += "Age must be 18 or above\n";

    if (qualification === "")
        msg += "Qualification is required\n";
```

```
    if (skills.length === 0)
        msg += "Select at least one skill\n";

    if (experience === "")
        msg += "Select experience\n";

    if (!terms)
        msg += "You must accept Terms & Conditions\n";

    if (msg !== "") {
        alert(msg);
        return false;
    }

    alert("Application Submitted Successfully!");
    return true;
}
</script>

</body>
</html>
```





WEEK 5 &6

1)

```
<!DOCTYPE html>
<!--hello.php-->
<html lang="en">
  <head>
    <title>My first php page</title>
  </head>
  <body>
    <?php
    function isPrime($nums){
        if($nums<2){
        return false;
      }
      for($j=2;$j<=sqrt($nums);$j++){
        if($nums%$j==0){
                return false;
         }
      }
      return true;
    }
    $sum=0;
    for($i=2;$i<=100;$i++){
        if(isPrime($i)){
        $sum+=$i;
      }
    }
    echo "<p>sum: $sum</p>";
    ?>
  </body>
</html>
```

Output:

sum: 1060

2)

```
<!DOCTYPE html>
<!--hello.php-->
<html lang="en">
  <head>
    <title>My first php page</title>
  </head>
  <body>
    <?php
    $capital=array("India"=>"Delhi","Japan"=>"Tokyo","Italy"=>"Rome");
    foreach($capital as $country=>$city){
        echo "<p>The capital of $country is $city</p>";
    }
    ?>
  </body>
</html>
```

Output:

The capital of India is Delhi

The capital of Japan is Tokyo

The capital of Italy is Rome

3)

```
<?php
function containsXas($states,&$statesArray){
   $contains="xas";
   foreach($states as $s){
      $temp=substr($s,-3);
      if(str_contains($temp,$contains)){
         echo "States with 'xas':$s";
         echo "<br>";
         array_push($statesArray,$s);
      }
   }
}
function startKEndS($states,&$statesArray)
{
  foreach($states as $s)
  {
   $temp  = substr($s,0)."".substr($s,-1);
   if((strcasecmp($temp[0],"k")==0) && (strcasecmp($temp[-1],"s")==0))
   {
     echo "States starting with k and ending with s: $s";
     echo "<br>";
     array_unshift($statesArray,$s);
   }
  }
}

function startMEndS($states,&$statesArray)
{
  foreach($states as $s)
  {
```

```
    $temp  = substr($s,0)."".substr($s,-1);
    if($temp[0] == "M" && $temp[-1] == "s")
    {
      echo "States starting with M and ending with s: $s";
      echo "<br>";
      array_unshift($statesArray,$s);
    }
  }
}
function endA($states,&$statesArray)
{
  foreach($states as $s)
  {
    $temp  = substr($s,-1);
    if($temp[-1] == "a")
    {
      echo "States ending with a: $s";
      echo "<br>";
      array_unshift($statesArray,$s);
    }
  }
}
function firstState($states,&$statesArray)
{
  $temp = $states[0];
  if($temp[0] == "M")
  {
    echo "States ending with a: $temp";
    echo "<br>";
    array_unshift($statesArray,$temp);
  }
```

```
}
function outArray($statesArray)
{
 echo "Printing states from Array : ";
 foreach($statesArray as $s)
 {
  echo "$s ";
 }
 echo "<br>";
}
$states = array("Mississippi","Alabama","Texas","Massachusetts","Kansas");
$statesArray = array();
containsXas($states,$statesArray);
startKEndS($states,$statesArray);
startMEndS($states,$statesArray);
endA($states,$statesArray);
firstState($states,$statesArray);
outArray($statesArray);
?>
```

Output:

States with 'xas':Texas
States starting with k and ending with s: Kansas
States starting with M and ending with s: Massachusetts
States ending with a: Alabama
States ending with a: Mississippi
Printing states from Array : Mississippi Alabama Massachusetts Kansas Texas

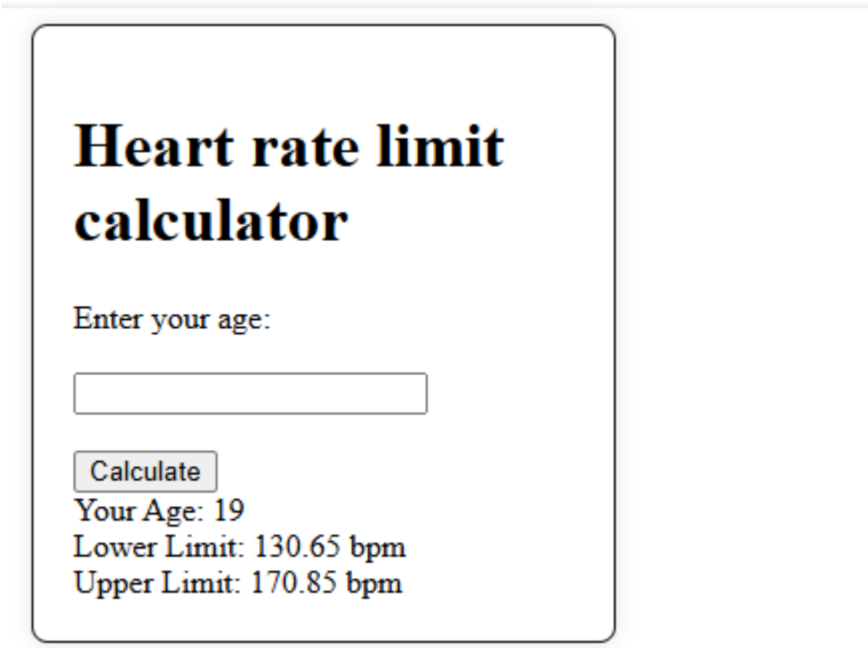
4)

```
<!DOCTYPE html>
<html>
  <head>
    <title>Heart rate limit calculator</title>
    <style>
```

```
        div{
            width: 250px;
            margin: auto;
            padding: 20px;
            border: 1px solid #111010;
            border-radius: 8px;
            box-shadow: 0 0 10px rgba(0,0,0,0.1);
        }
    </style>
</head>
<body>
    <div>
        <h1>Heart rate limit calculator</h1>
        <form method="post">
            Enter your age:<br><br>
            <input type="number" name="age" requiredmin="1" max="120">
            <br><br>
            <input type="submit" value="Calculate">
        </form>

        <?php
        if($_SERVER["REQUEST_METHOD"]=="POST"){
            $age=$_POST['age'];
            if($age>0){
                $upper=(220-$age)*0.85;
                $lower=(220-$age)*0.65;
                echo "Your Age: $age <br>";
                echo "Lower Limit: " . round($lower, 2) . " bpm<br>";
                echo "Upper Limit: " . round($upper, 2) . " bpm";


            }
```

```
            else {
                echo "<div class='result'>Please enter a valid age.</div>";
            }


        }
        ?>
    </div>
</body>
</html>
```

Output:



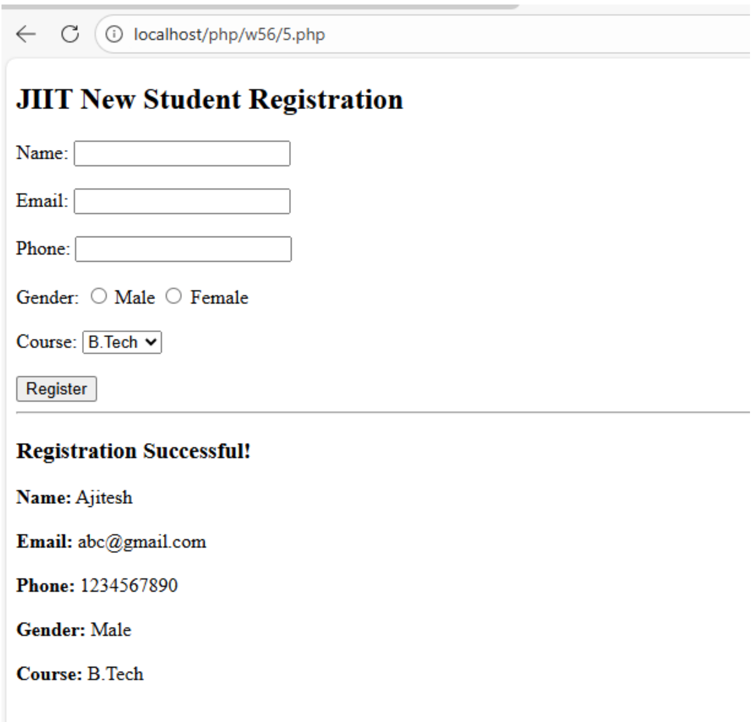
Q5

```
<!DOCTYPE html>
<html>
<head>
    <title>JIIT New Student Registration</title>


    <script>
```

```
        function validateForm() {

            var name = document.forms["regForm"]["name"].value;
            var email = document.forms["regForm"]["email"].value;
            var phone = document.forms["regForm"]["phone"].value;

            if (name == "") {
                alert("Name is required");
                return false;
            }

            var emailPattern = /^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$/;
            if (!emailPattern.test(email)) {
                alert("Enter valid Email Address");
                return false;
            }

            var phonePattern = /^[0-9]{10}$/;
            if (!phonePattern.test(phone)) {
                alert("Enter valid 10-digit phone number");
                return false;
            }

            return true;
        }
    </script>
</head>
<body>

<h2>JIIT New Student Registration</h2>
```

```
<form name="regForm" method="post" onsubmit="return validateForm()">

    Name: <input type="text" name="name"><br><br>

    Email: <input type="text" name="email"><br><br>

    Phone: <input type="text" name="phone"><br><br>

    Gender:
    <input type="radio" name="gender" value="Male"> Male
    <input type="radio" name="gender" value="Female"> Female
    <br><br>

    Course:
    <select name="course">
        <option value="B.Tech">B.Tech</option>
        <option value="M.Tech">M.Tech</option>
        <option value="MBA">MBA</option>
    </select>
    <br><br>

    <input type="submit" name="submit" value="Register">

</form>

<hr>

<?php
if (isset($_POST['submit'])) {

    $name   = $_POST['name'];
```

```
    $email  = $_POST['email'];
    $phone  = $_POST['phone'];
    $gender = $_POST['gender'];
    $course = $_POST['course'];

    echo "<h3>Registration Successful!</h3>";

    echo "<b>Name:</b> $name <br><br>";
    echo "<b>Email:</b> $email <br><br>";
    echo "<b>Phone:</b> $phone <br><br>";
    echo "<b>Gender:</b> $gender <br><br>";
    echo "<b>Course:</b> $course <br><br>";
}
?>

</body>
</html>
```

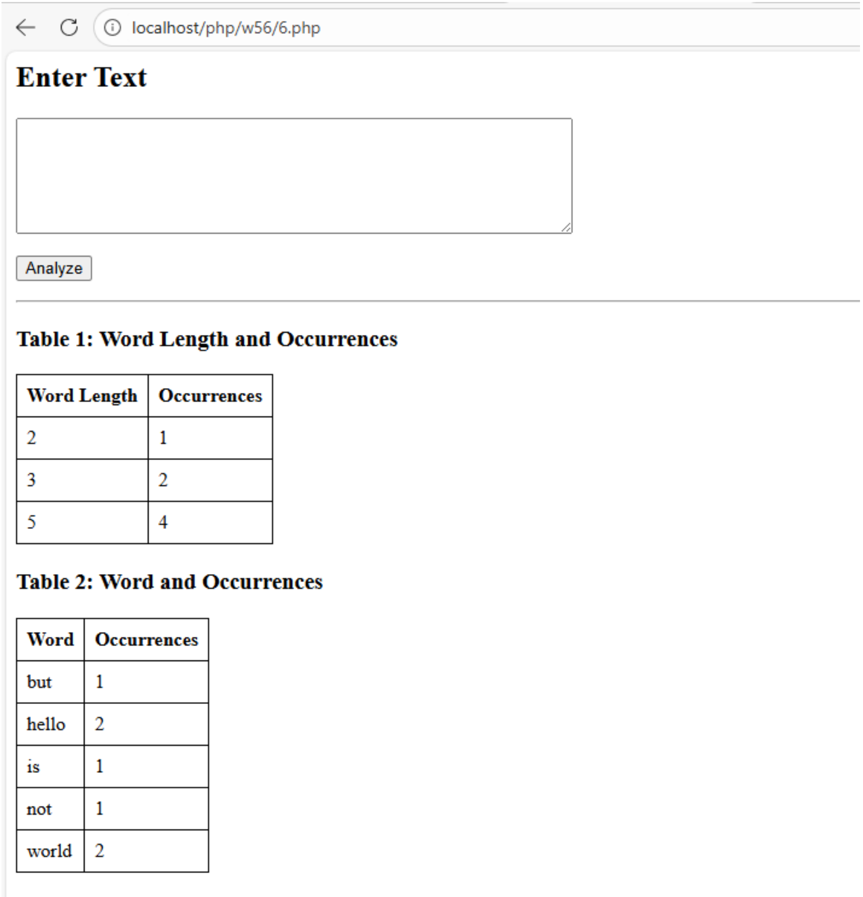


Q6

```
<html>
<head>
    <title>Word Length and Occurrence Counter</title>
    <style>
        table, th, td {
            border: 1px solid black;
            border-collapse: collapse;
            padding: 8px;
        }
        table {
            margin-bottom: 20px;
        }
    </style>
</head>
<body>

<h2>Enter Text</h2>

<form method="post">
    <textarea name="text" rows="6" cols="60" required></textarea><br><br>
    <input type="submit" name="submit" value="Analyze">
</form>

<hr>

<?php
if (isset($_POST['submit'])) {

    $text = $_POST['text'];
```

```
$text = strtolower($text);

$words = preg_split("/[^a-zA-Z]+/", $text);

$lengthCount = array();
$wordCount = array();

foreach ($words as $word) {

    if ($word != "") {

        $length = strlen($word);

        if (isset($lengthCount[$length]))
            $lengthCount[$length]++;
        else
            $lengthCount[$length] = 1;

        if (isset($wordCount[$word]))
            $wordCount[$word]++;
        else
            $wordCount[$word] = 1;
    }
}

ksort($lengthCount);
ksort($wordCount);

echo "<h3>Table 1: Word Length and Occurrences</h3>";
echo "<table>";
```

```
    echo "<tr><th>Word Length</th><th>Occurrences</th></tr>";

    foreach ($lengthCount as $length => $count) {
        echo "<tr><td>$length</td><td>$count</td></tr>";
    }

    echo "</table>";

    echo "<h3>Table 2: Word and Occurrences</h3>";
    echo "<table>";
    echo "<tr><th>Word</th><th>Occurrences</th></tr>";

    foreach ($wordCount as $word => $count) {
        echo "<tr><td>$word</td><td>$count</td></tr>";
    }

    echo "</table>";
}
?>

</body>
</html>
```



localhost/php/w56/6.php

Enter Text

Analyze

Table 1: Word Length and Occurrences

Word Length	Occurrences
2	1
3	2
5	4

Table 2: Word and Occurrences

Word	Occurrences
but	1
hello	2
is	1
not	1
world	2

Q7

```
<?php


$servername = "localhost";

$username = "root";

$password = "";


$conn = new mysqli($servername, $username, $password);


if ($conn->connect_error) {

   die("Connection failed: " . $conn->connect_error);

}
```

```
echo "<h3>Connected Successfully</h3>";

$conn->query("CREATE DATABASE IF NOT EXISTS Firstdb");
$conn->select_db("Firstdb");

$sql = "CREATE TABLE IF NOT EXISTS Student (
   Enrollment_no INT PRIMARY KEY,
   Name VARCHAR(50),
   Age INT,
   Course VARCHAR(50)
)";

$conn->query($sql);

function display($conn) {
   $result = $conn->query("SELECT * FROM Student");

   echo "<table border='1' cellpadding='8'>";
   echo "<tr>
         <th>Enrollment No</th>
         <th>Name</th>
         <th>Age</th>
         <th>Course</th>
        </tr>";

   while ($row = $result->fetch_assoc()) {
      echo "<tr>
            <td>{$row['Enrollment_no']}</td>
            <td>{$row['Name']}</td>
            <td>{$row['Age']}</td>
```

```
            <td>{$row['Course']}</td>
           </tr>";
    }

    echo "</table><br>";
}

$conn->query("DELETE FROM Student");

$conn->query("INSERT INTO Student VALUES
(101,'Ajitesh',20,'IT'),
(102,'Riya',23,'IT'),
(103,'Rahul',25,'ECE'),
(104,'Sneha',24,'CSE'),
(105,'Karan',26,'ME'),
(106,'Vedanshi',21,'IT'),
(107,'Vikas',27,'ECE'),
(108,'Neha',22,'CSE'),
(109,'Arjun',23,'IT'),
(110,'Simran',24,'CSE')");

echo "<h3>After Inserting 10 Students</h3>";
display($conn);

$conn->query("DELETE FROM Student WHERE Age > 24");

echo "<h3>After Deleting Students with Age > 24</h3>";
display($conn);

$conn->query("UPDATE Student SET Course='DBW' WHERE Enrollment_no=110");
```

```php
echo "<h3>After Updating Course of Enrollment_no 110 to DBW</h3>";
display($conn);

$conn->close();

?>
```

localhost/php/w56/7.php

Connected Successfully

After Inserting 10 Students

Enrollment No	Name	Age	Course
101	Ajitesh	20	IT
102	Riya	23	IT
103	Rahul	25	ECE
104	Sneha	24	CSE
105	Karan	26	ME
106	Vedanshi	21	IT
107	Vikas	27	ECE
108	Neha	22	CSE
109	Arjun	23	IT
110	Simran	24	CSE

After Deleting Students with Age > 24

Enrollment No	Name	Age	Course
101	Ajitesh	20	IT
102	Riya	23	IT
104	Sneha	24	CSE
106	Vedanshi	21	IT

localhost/php/w56/7.php

After Deleting Students with Age > 24

Enrollment No	Name	Age	Course
101	Ajitesh	20	IT
102	Riya	23	IT
104	Sneha	24	CSE
106	Vedanshi	21	IT
108	Neha	22	CSE
109	Arjun	23	IT
110	Simran	24	CSE

After Updating Course of Enrollment_no 110 to DBW

Enrollment No	Name	Age	Course
101	Ajitesh	20	IT
102	Riya	23	IT
104	Sneha	24	CSE
106	Vedanshi	21	IT
108	Neha	22	CSE
109	Arjun	23	IT
110	Simran	24	DBW

Q8

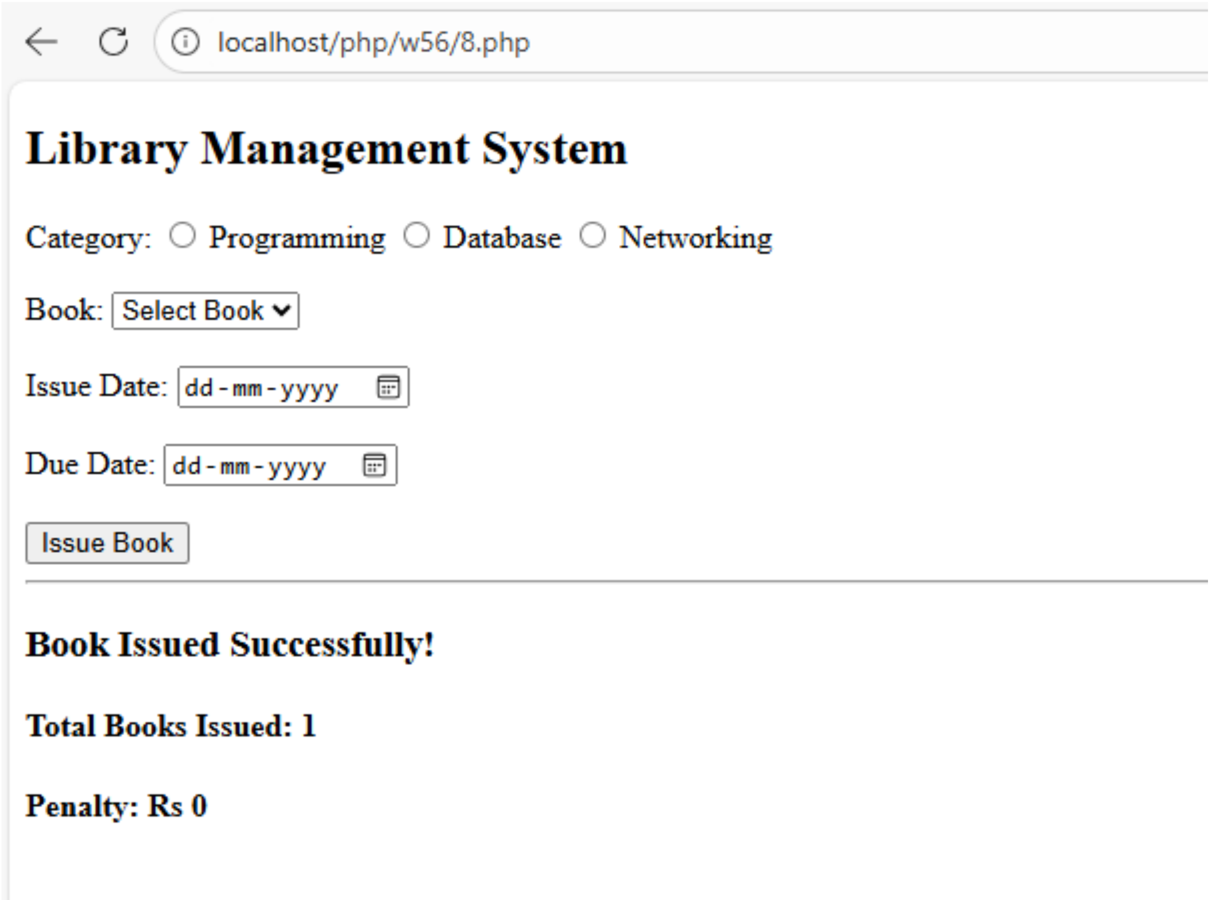
```
<?php

session_start();
```

```php
if (!isset($_SESSION['issued_count'])) {
    $_SESSION['issued_count'] = 0;
}

$categories = [
    "Programming" => ["C", "C++", "Java", "Python", "PHP"],
    "Database" => ["MySQL", "Oracle", "MongoDB", "SQL Server", "PostgreSQL"],
    "Networking" => ["TCP/IP", "CCNA", "Cyber Security", "Cloud", "Linux"]
];

$message = "";
$penalty = 0;

if (isset($_POST['issue'])) {

    $_SESSION['issued_count']++;

    $issue = new DateTime($_POST['issue_date']);
    $due = new DateTime($_POST['due_date']);

    if ($due < $issue) {
        $days = $issue->diff($due)->days;
        $penalty = $days * 10;
        $message = "Late return! Penalty Applied.";
    } else {
        $message = "Book Issued Successfully!";
    }
}
?>
```

```
<!DOCTYPE html>
<html>
<head>
  <title>Library System</title>

  <script>
    var books = {
      "Programming": ["C","C++","Java","Python","PHP"],
      "Database": ["MySQL","Oracle","MongoDB","SQL Server","PostgreSQL"],
      "Networking": ["TCP/IP","CCNA","Cyber Security","Cloud","Linux"]
    };

    function loadBooks(category) {
      var dropdown = document.getElementById("book");
      dropdown.innerHTML = "";

      books[category].forEach(function(book) {
        var option = document.createElement("option");
        option.value = book;
        option.text = book;
        dropdown.add(option);
      });
    }
  </script>
</head>
<body>

<h2>Library Management System</h2>

<form method="post">
```

```html
Category:
<input type="radio" name="category" value="Programming" onclick="loadBooks(this.value)" required> Programming
<input type="radio" name="category" value="Database" onclick="loadBooks(this.value)"> Database
<input type="radio" name="category" value="Networking" onclick="loadBooks(this.value)"> Networking

<br><br>

Book:
<select name="book" id="book" required>
  <option value="">Select Book</option>
</select>

<br><br>

Issue Date:
<input type="date" name="issue_date" required><br><br>

Due Date:
<input type="date" name="due_date" required><br><br>

<input type="submit" name="issue" value="Issue Book">

</form>

<hr>

<h3><?php echo $message; ?></h3>

<h4>Total Books Issued: <?php echo $_SESSION['issued_count']; ?></h4>
```

```
<h4>Penalty: Rs <?php echo $penalty; ?></h4>

</body>
</html>
```



WebTech

WEEK 7

992401040105 Ananya Soni

Q1

```
const{ parentPort } = require('worker_threads');
const sharp = require('sharp');

parentPort.on('message', async (filePath) => {
  await sharp(filePath).resize(800).toFile('resized.jpg');
  parentPort.postMessage('done');
});

// main.js
const{ Worker } = require('worker_threads');
constnodemailer = require('nodemailer');

function resizeImage(filePath) {
  return new Promise((resolve) => {
    const worker = new Worker('./image-resize.js');
    worker.postMessage(filePath);
    worker.on('message', resolve);
  });
}

async function sendEmail() {
  let transporter = nodemailer.createTransport({ sendmail: true });
  await transporter.sendMail({ from: 'me@test.com', to: 'user@test.com', subject: 'Success',
text: 'Image resized!' });
}

(async () => {
  await resizeImage('big.jpg'); // CPU-bound → Worker Thread
  await sendEmail(); // I/O-bound → Event Loop
})();
```

Q2

```
const fs = require('fs');
const{ pipeline } = require('stream');
const{ promisify } = require('util');
constpipelineAsync = promisify(pipeline);

async function copyFile() {
  const source = fs.createReadStream('fast-ssd-file.txt', { highWaterMark: 64 * 1024 });
  const destination = fs.createWriteStream('cloud-storage.txt');
```

```
  try {
    await pipelineAsync(source, destination);
    console.log('Copy complete without memory spike!');
  } catch (err) {
    console.error('Pipeline failed:', err);
  }
}

copyFile();
```

Q3

```
let clicks = [];
document.addEventListener('click', (e) =>clicks.push(e));

constclicksMap = new WeakMap();

function trackClick(element) {
  element.addEventListener('click', () => {
    clicksMap.set(element, Date.now());
  }, { once: true }); // auto-remove listener
}

const controller = new AbortController();
document.addEventListener('click', handler, { signal: controller.signal });

controller.abort();
```

Q4

```
interface User {
  id: number;
  name: string;
}
const user: User = { id: 1, name: 'Shaurya' };

// But runtime constructs like enums/decorators need transpilation
enum Role { Admin, User }
console.log(Role.Admin); // Needs emitted JS, not just type stripping

function Log(target: any) {
  console.log('Decorator applied to:', target);
}

@Log
class Test {}
```

Q5

```
const express = require('express');
const app = express();
const server = app.listen(3000);

app.get('/', (req, res) =>res.send('Hello'));

process.on('SIGTERM', () => {
  console.log('SIGTERM received. Closing server...');
  server.close(() => {
    console.log('All requests finished. Server closed.');
    process.exit(0);
  });
});
```

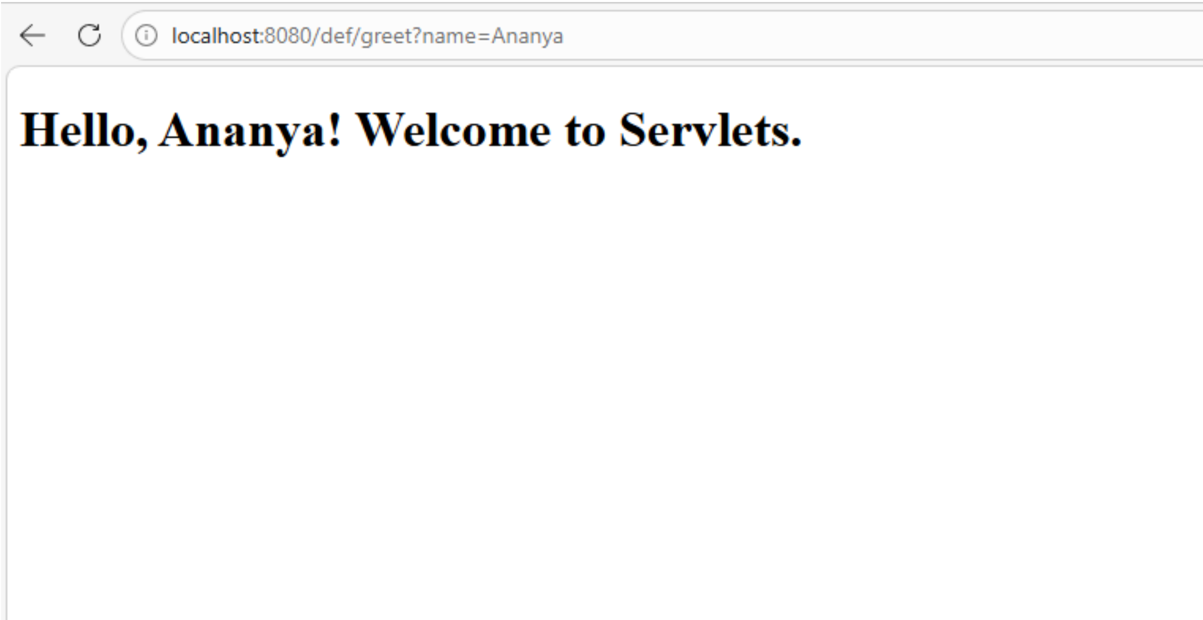
WEEK 9

ANANYA SONI 992401040105

1)

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet("/greet")
public class uifegw extends HttpServlet {
    private static final long serialVersionUID = 1L;

    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        String name = request.getParameter("name");


        if (name == null || name.trim().isEmpty()) {
            name = "Guest";
        }


        try (PrintWriter out = response.getWriter()) {
            out.println("<!DOCTYPE html>");
            out.println("<html>");
            out.println("<head><title>Greeting Servlet</title></head>");
            out.println("<body>");
            out.printf("<h1>Hello, %s! Welcome to Servlets.</h1>%n",
escapeHtml(name));
            out.println("</body>");
            out.println("</html>");
        }
    }
  private String escapeHtml(String input) {
        return input.replace("&", "&amp;")
                    .replace("<", "&lt;")
                    .replace(">", "&gt;")
                    .replace("\"", "&quot;")
                    .replace("'", "&#39;");
    }
}
```

Output:



2)

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

/**
 * A simple servlet to demonstrate the lifecycle methods:
 * init(), service(), and destroy().
 */
@WebServlet("/LifecycleDemo")
public class uifegw extends HttpServlet {

    // Called once when the servlet is first loaded into memory
    @Override
    public void init() throws ServletException {
        super.init();
        System.out.println("Servlet init() method called - Servlet is being
initialized.");
    }

    // Called for every request to the servlet
    @Override
    protected void service(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        System.out.println("Servlet service() method called - Processing
request.");

        // Set response content type
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        // Output to browser
        out.println("<html><body>");
```

```
        out.println("<h2>Servlet Lifecycle Demonstration</h2>");
        out.println("<p><b>init()</b> was called when the servlet was first 
loaded.</p>");
        out.println("<p><b>service()</b> is called for every request.</p>");
        out.println("<p>Check server logs to see the execution order.</p>");
        out.println("</body></html>");
    }

    // Called once when the servlet is being removed from memory
    @Override
    public void destroy() {
        System.out.println("Servlet destroy() method called - Servlet is being 
destroyed.");
    }
}
```

Output:

```
Servlet init() method called - Servlet is being initialized.
Servlet service() method called - Processing request.
Servlet service() method called - Processing request.
```

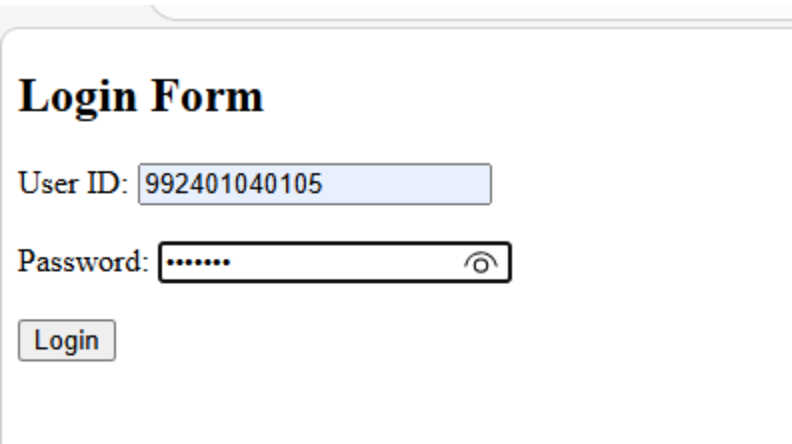
```
Apr 23, 2026 11:35:02 AM org.apache.catalina.core.StandardService stopInternal
INFO: Stopping service [Catalina]
Servlet destroy() method called - Servlet is being destroyed.
```

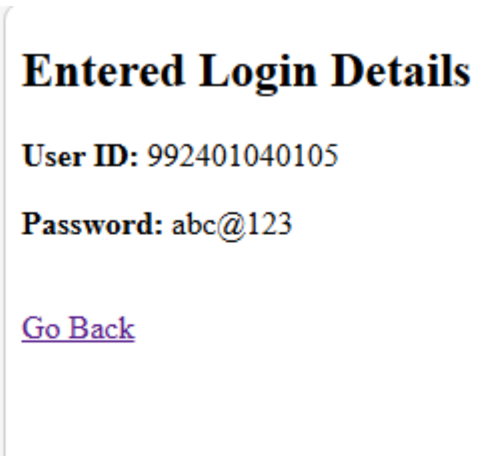
3)

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet("/LoginServlet")
public class uifegw extends HttpServlet {
    private static final long serialVersionUID = 1L;
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        out.println("<!DOCTYPE html>");
        out.println("<html><head><title>Login Form</title></head><body>");
        out.println("<h2>Login Form</h2>");
        out.println("<form method='post' action='LoginServlet'>");
        out.println("User ID: <input type='text' name='userid' 
required><br><br>");
        out.println("Password: <input type='password' name='password' 
required><br><br>");
        out.println("<input type='submit' value='Login'>");
        out.println("</form>");
        out.println("</body></html>");
    }

    @Override
```

```
    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        String userId = request.getParameter("userid");
        String password = request.getParameter("password");

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        out.println("<!DOCTYPE html>");
        out.println("<html><head><title>Login Result</title></head><body>");
        out.println("<h2>Entered Login Details</h2>");
        out.println("<p><strong>User ID:</strong> " + userId + "</p>");
        out.println("<p><strong>Password:</strong> " + password + "</p>");
        out.println("<br><a href='LoginServlet'>Go Back</a>");
        out.println("</body></html>");
    }
}
```

Output:





4)

import java.io.IOException;

import java.io.PrintWriter;

import java.sql.Connection;

import java.sql.DriverManager;

```java
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.Statement;

import javax.servlet.ServletConfig;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet("/newservlet")
public class newservlet extends HttpServlet {

  private static final long serialVersionUID = 1L;

  public newservlet() {
    super();
  }

  public void init(ServletConfig config) throws ServletException {
    System.out.println("Servlet initialized (init() called)");
  }

  public void service(HttpServletRequest request,
            HttpServletResponse response)
      throws ServletException, IOException {

    System.out.println("Service method is called");

    super.service(request, response);
```

```
}

public void destroy() {
   System.out.println("Servlet destroyed (destroy() called)");
}

protected void doGet(HttpServletRequest request,
                 HttpServletResponse response)
      throws ServletException, IOException {

   doPost(request, response);
}

protected void doPost(HttpServletRequest request,
                  HttpServletResponse response)
      throws ServletException, IOException {

   response.setContentType("text/html");

   PrintWriter out = response.getWriter();

   Connection con = null;
   Statement stmt = null;
   PreparedStatement ps = null;
   ResultSet rs = null;

   try {

      Class.forName("com.mysql.cj.jdbc.Driver");

      con = DriverManager.getConnection(
```

```
        "jdbc:mysql://localhost:3306/test",
        "root",
        "root"
);

stmt = con.createStatement();

stmt.executeUpdate("DELETE FROM Student");


ps = con.prepareStatement(
        "INSERT INTO Student VALUES (?, ?, ?, ?, ?, ?)"
);



ps.setInt(1, 1);
ps.setString(2, "Aman");
ps.setInt(3, 20);
ps.setInt(4, 2);
ps.setDouble(5, 8.5);
ps.setString(6, "CSE");
ps.addBatch();



ps.setInt(1, 2);
ps.setString(2, "Riya");
ps.setInt(3, 19);
ps.setInt(4, 1);
ps.setDouble(5, 9.1);
ps.setString(6, "ECE");
ps.addBatch();
```

```java
ps.setInt(1, 3);
ps.setString(2, "Priya");
ps.setInt(3, 18);
ps.setInt(4, 1);
ps.setDouble(5, 9.2);
ps.setString(6, "CSE");
ps.addBatch();

ps.setInt(1, 4);
ps.setString(2, "Shriya");
ps.setInt(3, 21);
ps.setInt(4, 3);
ps.setDouble(5, 8.8);
ps.setString(6, "IT");
ps.addBatch();

ps.setInt(1, 5);
ps.setString(2, "Liya");
ps.setInt(3, 17);
ps.setInt(4, 1);
ps.setDouble(5, 8.3);
ps.setString(6, "ECE");
ps.addBatch();

ps.setInt(1, 6);
ps.setString(2, "Kiara");
ps.setInt(3, 20);
ps.setInt(4, 2);
ps.setDouble(5, 9.3);
ps.setString(6, "ECE");
```

```
ps.addBatch();

ps.executeBatch();

out.println("<h3>Batch Insert Successful</h3>");


ps = con.prepareStatement(
    "DELETE FROM Dept WHERE numphds < ?"
);

ps.setInt(1, 2);

int rows = ps.executeUpdate();

out.println("<h3>" + rows +
    " rows deleted from Dept table</h3>");


out.println("<h2>Student Table</h2>");

rs = stmt.executeQuery("SELECT * FROM Student");

while(rs.next()) {

  out.println(
      rs.getInt("sid") + " " +
      rs.getString("sname") + " " +
      rs.getInt("age") + " " +
      rs.getInt("year") + " " +
      rs.getDouble("cgpa") + " " +
```

```
        rs.getString("dname") +
        "<br>"
    );
}

out.println("<h2>Dept Table</h2>");

rs = stmt.executeQuery("SELECT * FROM Dept");

while(rs.next()) {

    out.println(
        rs.getString("dname") + " " +
        rs.getInt("numphds") +
        "<br>"
    );
}


out.println("<h2>Prof Table</h2>");

rs = stmt.executeQuery("SELECT * FROM Prof");

while(rs.next()) {

    out.println(
        rs.getString("pname") + " " +
        rs.getString("dname") +
        "<br>"
    );
}
```

```
out.println("<h2>Course Table</h2>");

rs = stmt.executeQuery("SELECT * FROM Course");

while(rs.next()) {

   out.println(
         rs.getInt("cno") + " " +
         rs.getString("cname") + " " +
         rs.getString("dname") +
         "<br>"
   );
}


out.println("<h2>CSE Students</h2>");

ps = con.prepareStatement(
      "SELECT sid, sname FROM Student WHERE dname = ?"
);

ps.setString(1, "CSE");

rs = ps.executeQuery();

while(rs.next()) {

   out.println(
         rs.getInt("sid") + " " +
```

```
                rs.getString("sname") +
                "<br>"
        );
    }

}

catch(Exception e) {

    out.println("<h3>Error:</h3>");
    out.println(e);
}

finally {

    try {

        if(rs != null)
            rs.close();

        if(stmt != null)
            stmt.close();

        if(ps != null)
            ps.close();

        if(con != null)
            con.close();

    }
```

```
        catch(Exception e) {

            out.println(e);
        }
      }
   }
}
```

Output:

Batch Insert Successful

0 rows deleted from Dept table

Student Table

1 Aman 20 2 8.5 CSE
2 Riya 19 1 9.1 ECE
3 Priya 18 1 9.2 CSE
4 Shriya 21 3 8.8 IT
5 Liya 17 1 8.3 ECE
6 Kiara 20 2 9.3 ECE

Dept Table

CSE 5
ECE 3

Prof Table

Dr. Sharma CSE
Dr. Mehta ECE
Dr. Rao IT

Course Table

101 DBMS CSE
102 Networks ECE
103 OS CSE
104 AI CSE

CSE Students

1 Aman
3 Priya

5)

```
<!DOCTYPE html>

<html>

<head>
```

```
<title>Simple Form</title>
</head>
<body>
<h2>Enter Details</h2>
<form action="MyServlet" method="post">
    UserID:<input type="text" name="name"><br><br>
    Age:<input type="number" name="age"><br><br>
    <input type="submit" value="Submit">
</form>
</body>
</html>
import java.io.*;
import javax.servlet.*;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;
import java.sql.*;

@WebServlet("/MyServlet")
public class MyServlet extends HttpServlet {

   protected void doPost(HttpServletRequest request, HttpServletResponse response)
         throws ServletException, IOException {

      response.setContentType("text/html");
      PrintWriter out = response.getWriter();
      String name = request.getParameter("name");
      String age = request.getParameter("age");

      try {

         Class.forName("com.mysql.cj.jdbc.Driver");
```

```
            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/test", "root", "root");


            PreparedStatement ps = con.prepareStatement(
                "INSERT INTO users(name, age) VALUES(?, ?)");

            ps.setString(1, name);
            ps.setInt(2, Integer.parseInt(age));
            ps.executeUpdate();
            out.println("<h2>Data inserted successfully!</h2>");

            con.close();

        } catch (Exception e) {
            e.printStackTrace();
            out.println("Error: " + e.getMessage());
        }
    }
}
```

Output:

```
mysql> select * from users;
+----+--------+------+
| id | name   | age  |
+----+--------+------+
|  1 | Ananya |   12 |
|  2 | Ananya |   12 |
+----+--------+------+
2 rows in set (0.00 sec)
```

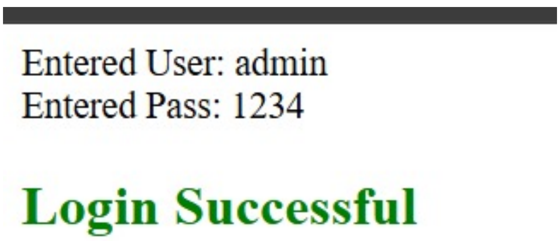
Data inserted successfully!

6)

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;
import java.sql.*;

@WebServlet("/loginservlet")
public class loginservlet extends HttpServlet {

  protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

    response.setContentType("text/html");
    PrintWriter out = response.getWriter();

    String userid = request.getParameter("name");
    String password = request.getParameter("pass");

    try {
      Class.forName("com.mysql.cj.jdbc.Driver");


      Connection con = DriverManager.getConnection(
        "jdbc:mysql://localhost:3306/test", "root", "root");
```

```
            PreparedStatement ps = con.prepareStatement(
                "SELECT * FROM usermaster WHERE userid=? AND password=?");

            ps.setString(1, userid);
            ps.setString(2, password);

            ResultSet rs = ps.executeQuery();
            out.println("Entered User: " + userid + "<br>");
            out.println("Entered Pass: " + password + "<br>");

            if (rs.next()) {
                out.println("<h2 style='color:green'>Login Successful</h2>");
            } else {
                out.println("<h2 style='color:red'>Invalid User ID or Password</h2>");
            }
            rs.close();
            ps.close();
            con.close();

        } catch (Exception e) {
            out.println(e);
        }
    }
}
```

Output:



WEEK 10

ANANYA SONI 992401040105

1)

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Insert title here</title>
</head>
<body>
<%
int a=10;
int b=8;
%>
<%!
public int multiply(int a,int b){
	return a*b;
}
%>
<b>SUM:</b>
<%=
a+b
%>
<p>
<b>MULTIPLICATION: </b><%=multiply(a,b) %>
</p>
</body>
</html>
```

Output:

2)

Header.jsp

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Insert title here</title>
</head>
<body>
<h1>JSP Q2</h1>
<p>DEMO FILE</p>
```

```
</body>
</html>
```

Footer.jsp

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Insert title here</title>
</head>
<body>
<p>&copy;<%=java.time.Year.now() %>JSP FILE COPYRIGHT</p>
</body>
</html>
```

Main.jsp

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Insert title here</title>
</head>
<body>
<%@ include file="header.jsp" %>
<p>Date: <%=new java.util.Date() %></p>
<%@ include file="footer.jsp" %>
</body>
</html>
```

Output:

3)

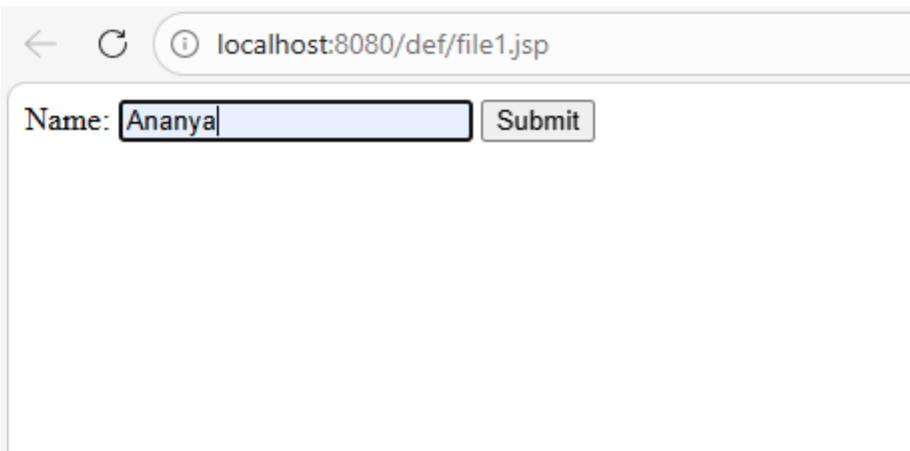
File1.jsp

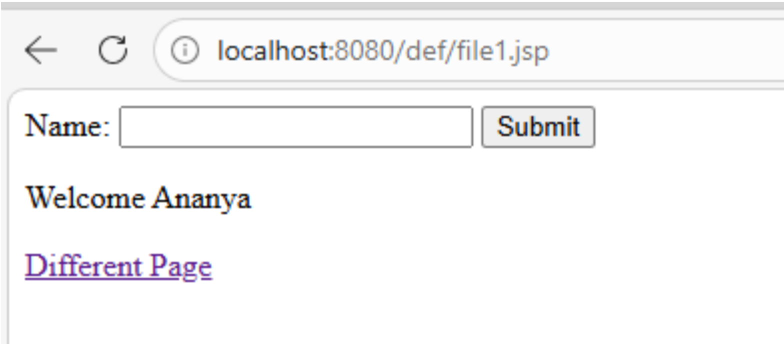
```
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Insert title here</title>
</head>
<body>
<form method="post" action="file1.jsp">
Name: <input type="text" name="username" required>
<input type="submit" name="submit">
</form>
<%
String name=request.getParameter("username");
session.setAttribute("user", name);
if(name!=null){
	out.println("<p>Welcome "+name+"</p>");
	out.println("<a href='file2.jsp'>Different Page</a>");
}
%>

</body>
</html>
```

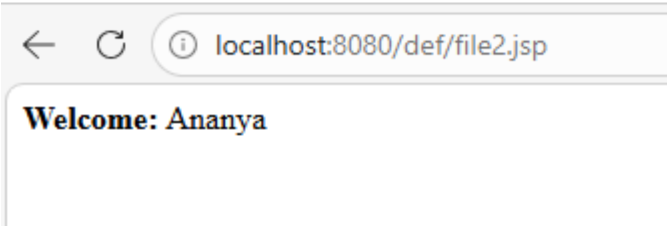
File2.jsp

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Insert title here</title>
</head>
<body>
<b>Welcome: </b> <%= session.getAttribute("user") %>
</body>
</html>
```

Output:







4)

```
<%@ page import="java.sql.*" %>

<html>
<head>
   <title>Q4</title>
</head>
<body>

<h2>Student Form</h2>

<form method="post">

   Name:
   <input type="text" name="name" required>
   <br><br>

   Marks:
   <input type="number" name="marks" required>
   <br><br>
```

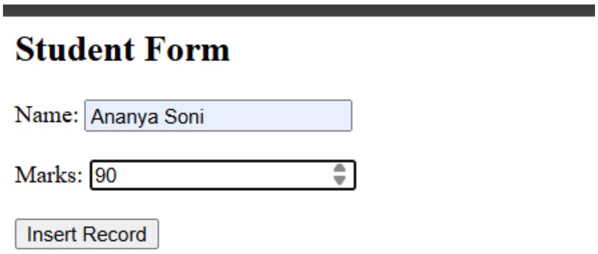
```
    <input type="submit" value="Insert Record">

</form>

<%



    String name = request.getParameter("name");
    String marksStr = request.getParameter("marks");


    if(name != null && marksStr != null)
    {

        Connection con = null;
        PreparedStatement ps = null;

        try
        {

            int marks = Integer.parseInt(marksStr);
Ξ

            Class.forName("com.mysql.cj.jdbc.Driver");


            con = DriverManager.getConnection(
                    "jdbc:mysql://localhost:3306/test",
                    "root",
                    "root"
```

```
        );

        String query =
            "INSERT INTO student(name, marks) VALUES(?, ?)";

        ps = con.prepareStatement(query);

        ps.setString(1, name);
        ps.setInt(2, marks);

        int rows = ps.executeUpdate();

        if(rows > 0)
        {
%>

          <h3 style="color:green;">
            Record inserted successfully!
          </h3>

<%
        }
        else
        {
%>

          <h3 style="color:red;">
            Record insertion failed!
          </h3>

<%
```

```
        }

    }

    catch(Exception e)
    {
%>

        <h3 style="color:red;">
            Error: <%= e %>
        </h3>

<%
    }

    finally
    {

        try
        {

            if(ps != null)
                ps.close();

            if(con != null)
                con.close();

        }

        catch(Exception e)
        {
```

```
            out.println(e);
          }
        }
    }

%>

</body>
</html>
```

Output:



5)

Login.jsp

```
<%@ page import="java.sql.*" %>

<html>
<head>
   <title>q5</title>
```

```
</head>
<body>

<h2>User Login</h2>

<form method="post">

   Username:
   <input type="text" name="username" required>
   <br><br>

   Password:
   <input type="password" name="password" required>
   <br><br>

   <input type="submit" value="Login">

</form>

<%

   String username = request.getParameter("username");
   String password = request.getParameter("password");

   if(username != null && password != null)
   {

      Connection con = null;
      PreparedStatement ps = null;
      ResultSet rs = null;
```

```
try
{

    Class.forName("com.mysql.cj.jdbc.Driver");

    con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/loginDB",
            "root",
            "root"
    );

    String query =
        "SELECT * FROM users WHERE username=? AND password=?";

    ps = con.prepareStatement(query);

    ps.setString(1, username);
    ps.setString(2, password);

    rs = ps.executeQuery();

    if(rs.next())
    {
```

```
            session.setAttribute("user", username);



            response.sendRedirect("dashboard.jsp");
          }
          else
          {
%>

            <h3 style="color:red;">
              Invalid Username or Password
            </h3>

<%
          }

        }

        catch(Exception e)
        {
          out.println(e);
        }

        finally
        {

          try
          {

            if(rs != null)
              rs.close();
```

```
            if(ps != null)
                ps.close();

            if(con != null)
                con.close();

        }

        catch(Exception e)
        {
            out.println(e);
        }
    }
}

%>

</body>
</html>
```

Dashboard.jsp

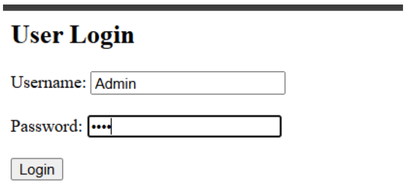
```
<%

    String user = (String)session.getAttribute("user");

    if(user == null)
    {
        response.sendRedirect("login.jsp");
    }

%>
```

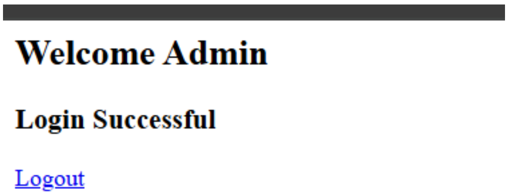
```
<html>
<head>
   <title>q5.2</title>
</head>
<body>

<h2>
   Welcome <%= user %>
</h2>

<h3>
   Login Successful
</h3>

<a href="logout.jsp">
   Logout
</a>

</body>
</html>
```

Logout.jsp

```
<%

   session.invalidate();


   response.sendRedirect("login.jsp");

%>
```

Output:





6)

Index.jsp

```
<%@ page import="java.sql.*" %>

<%@ include file="header.jsp" %>

<%!

   Connection con = null;

   public Connection getConnection() throws Exception
```

```
  {
     Class.forName("com.mysql.cj.jdbc.Driver");

     con = DriverManager.getConnection(
           "jdbc:mysql://localhost:3306/crudDB",
           "root",
           "root"
     );

     return con;
  }

%>

<html>
<head>

<title>CRUD Application</title>

<style>

table, th, td{
  border:1px solid black;
  border-collapse:collapse;
  padding:10px;
}

</style>

</head>
```

```
<body>

<h2>Add Student Record</h2>

<form method="post">

    Name:
    <input type="text" name="name" required>

    <br><br>

    Marks:
    <input type="number" name="marks" required>

    <br><br>

    <input type="submit" name="add" value="Add Record">

</form>

<br>

<%

    String name = request.getParameter("name");
    String marksStr = request.getParameter("marks");

    if(request.getParameter("add") != null)
    {

        try
```

```
{

    int marks = Integer.parseInt(marksStr);

    con = getConnection();

    PreparedStatement ps =
        con.prepareStatement(
            "INSERT INTO student(name, marks) VALUES(?, ?)"
        );

    ps.setString(1, name);
    ps.setInt(2, marks);

    int rows = ps.executeUpdate();

    if(rows > 0)
    {

        session.setAttribute(
            "msg",
            "Record Inserted Successfully"
        );

        out.println(
            "<h3 style='color:green;'>"
            + session.getAttribute("msg")
            + "</h3>"
        );
    }
```

```
            ps.close();
            con.close();

        }

        catch(Exception e)
        {
            out.println(e);
        }
    }

%>

<%



    String deleteId = request.getParameter("delete");

    if(deleteId != null)
    {

        try
        {

            con = getConnection();

            PreparedStatement ps =
                con.prepareStatement(
                    "DELETE FROM student WHERE id=?"
                );
```

```
            ps.setInt(1, Integer.parseInt(deleteId));

            ps.executeUpdate();

            response.sendRedirect("index.jsp");

            ps.close();
            con.close();

        }

        catch(Exception e)
        {
            out.println(e);
        }
    }

%>

<h2>Student Records</h2>

<table>

<tr>

    <th>ID</th>
    <th>Name</th>
    <th>Marks</th>
    <th>Action</th>
```

```
</tr>

<%



   try
   {

      con = getConnection();

      Statement st = con.createStatement();

      ResultSet rs =
         st.executeQuery(
            "SELECT * FROM student"
         );

      while(rs.next())
      {

%>

<tr>

   <td>
      <%= rs.getInt("id") %>
   </td>

   <td>
      <%= rs.getString("name") %>
```

```
    </td>

    <td>
        <%= rs.getInt("marks") %>
    </td>

    <td>

        <a href="index.jsp?delete=<%= rs.getInt("id") %>">
            Delete
        </a>

    </td>

</tr>

<%

        }

        rs.close();
        st.close();
        con.close();

    }

    catch(Exception e)
    {
        out.println(e);
    }
```

```
%>

</table>

</body>
</html>

<%@ include file="footer.jsp" %>
```

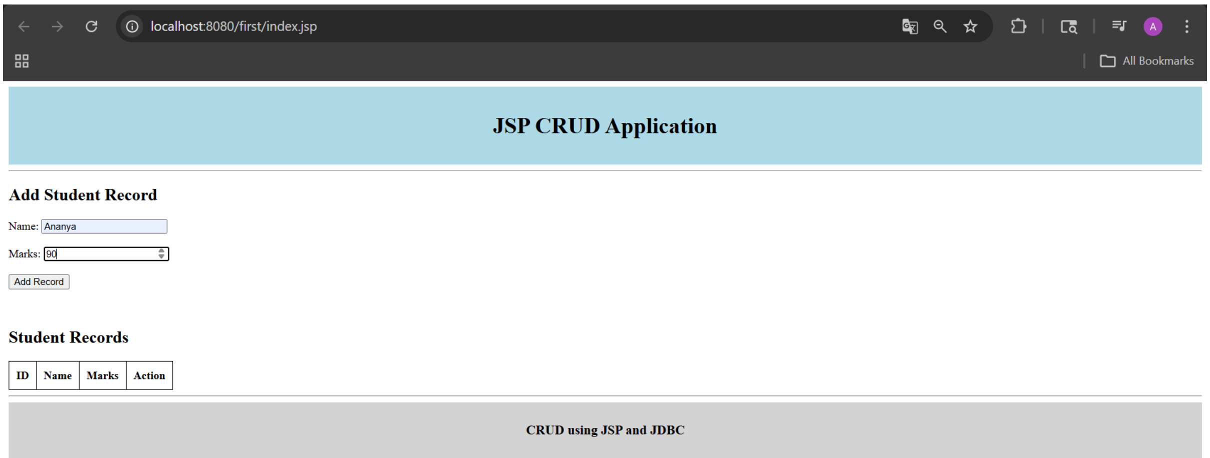
Header.jsp

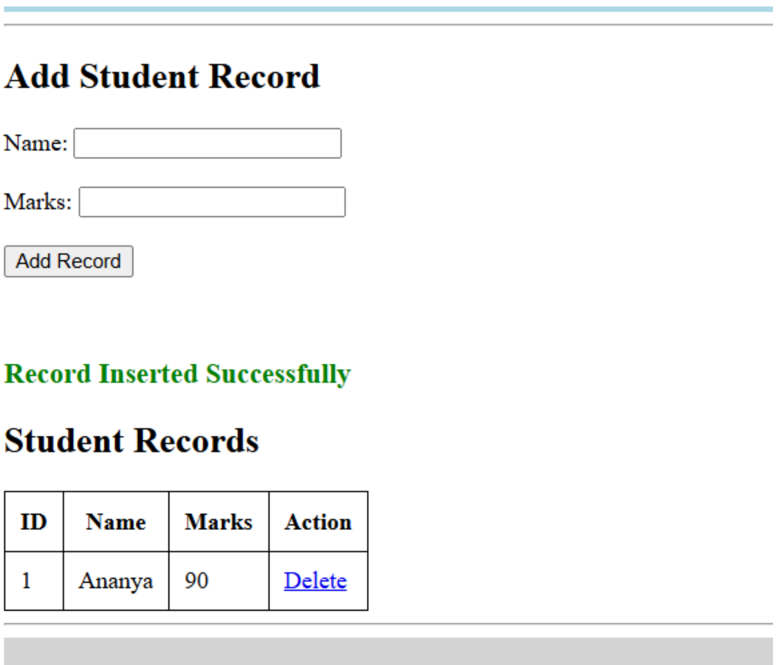
```
<div style="background-color:lightblue;padding:15px;">
  <h1 align="center">
    JSP CRUD Application
  </h1>
</div>

<hr>
```

Footer.jsp

```
<hr>

<div style="background-color:lightgray;padding:10px;">
  <h3 align="center">
    CRUD using JSP and JDBC
  </h3>
</div>
```

Output:


localhost:8080/first/index.jsp
All Bookmarks
JSP CRUD Application
Add Student Record
Name: Ananya
Marks: 90
Add Record
Student Records
ID
Name
Marks
Action
CRUD using JSP and JDBC


Add Student Record
Name:
Marks:
Add Record
Record Inserted Successfully
Student Records
ID
Name
Marks
Action
1
Ananya
90
Delete